ActInf Livestream #027.2 ~ “Active Inference: Applicability to Different Types of ..."

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hello everyone and welcome to the active inference lab this is active lab live stream number 27.2 on august 31st 2021 we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll follow video etiquette for live streams and see who joins us the short link is the calendar of live streams of the past and future and we did choose paper but didn't update this slide so check the link for next week's paper today in 27.2 welcome to stephen greetings in 27.2 we're joined by the two stevens as it were and um we'll have a few questions that we can have slides for the figures to go through the paper itself anything else that anyone wants to add and then of course any questions that people type into live chat as well if they're watching live so um we can just say hello and then we can [Music] leave it to the author last just to maybe open it up or just bring what they were thinking about our second discussion here today so i'm daniel and i'm a postdoc in california i'll pass to stephen celet oh hello there i'm stephen i'm based in toronto and uh i'm working on social topographies and other spatial meaning making and hoping to find ways for active influence to shed light and all of that and i'll pass it shall pass it to the other stephen is that okay yeah good day to you i'm uh i'm in finland southern finland i'm based at the technical research center of finland and i'm ready to answer questions cool so again uh anyone who is watching can post a question but then there's a few that we have lined up a little bit just one theme that stood out as far as selecting this paper was that it was applied it was applied to making the kinds of objects and supporting the ability to have the internet and computers as well as just objects like pens and things like that so how does this idea of helping us design physical systems which ends up being our niche modification uh activity as an economy how do we help design maybe if we're not an industrial engineer but potentially people working on different kinds of projects that end up modifying the environment maybe even influencing the same systems that industrial engineers use or um using some of the same techniques but people who are just thinking along these lines and want to take what you translated one step towards application in your paper and then think about how it applies in their own work so cyber physical system so the latest um trend in this is um for example the in in the usa i think it's called smart manufacturing and in europe industry 4.0 and a big thing in this is the idea of having digital twins or digital siblings or even avatars um and the idea of this is is that there's a the physical machine the physical process has got um sensors on it and actuators on it and the sensors bring live data to the digital equivalent of this physical process and through so uh and then it seeks to optimize by seeing what's going on so and a basic basic level is although difficult to achieve is predictive maintenance for example so if nothing else at least to stop the physical process from malfunctioning or even worse stopping and there's no um there's no theoretical foundation for for these industrial innovations for things like digital twins so of course there's a theoretical and background of cyber physical systems but there's nothing to anchor it with in my opinion that's very straightforward so if we look at life sciences generally we know that evolution has had millennia of and millions of years to iterate to develop elegant solutions so it seems sensible at least to me to to look to the life sciences to see how nature's evolved towards least action and of course uh free energy principle and active inference framework or theory from the life sciences are in their essence i believe very as one would expect guiding principles in nature to be and applicable to human and artificial intelligence to artificial agents i believe it's useful to start from that basis and with the free energy principle this very straightforward idea of that of trying to minimize the information gap between the generative model and the world itself so that's a very straight forward idea and i know there's um you know the the mass behind it is evolving all the time and there's some debate and courteous dispute about that but the basic ideas seem straightforward so i believe it's therefore a good place to start because this is such a basic idea isn't it minimizing the information gap between the the generative model and the world itself is this what i've said helpful in any way thanks for this i wrote a lot of it down stephen's let yeah some really good points there one thing i'd be interested in to to know what your thoughts are is you know we can end up kind of adapting our niche through action as a kind of consequence of like following a path and these types of things and then we've got this kind of more constructed um sort of uh deliberative um organizational work um in which you know the in many ways the point of that work can be to construct the environment um in which tools will operate you know so in some ways we we use the tools of our legs or the tools of being in in the world and it can sculpt that niche and also we can be very much sculpting a bounded niche so i'm wondering my question is thinking about whether you see those that kind of distinction as being significant and how you see um the idea of like the free energy or the implications of the free energy sort of rolling out in the niche versus the generative model for that niche being embodied in some way so you know so anyway so how much is there something when it's embodied for instance when you're talking about these um total quality management environments where someone is um trying to maintain i don't know iso standards of manufacturing or whatever so i wonder what your thoughts are about that whether there's distinctions there in terms of active influence being particularly involved or you know the free energy principle more more generally so um so i believe that um that that the core of quality management is just a an instantiation of free uh of realizing the free energy principle through active inference and that's what i've you know put in the um but there's a bigger issue about um that in in nature things are evolving and uh as i put in another one of my papers in entropy that uh that there are no things are evolving to some degree naturally and so if you say predator and prey they they just it's evolved and if you're a beaver trying to get back to lodge that you've built your niche construction as a beaver and there's a links in if you're if you see in between where you are and you'll be the lodge and there's a links a hungry link standing in your way then it's clear that okay and this is a predator and i should this is not my as a as a beaver that's not my preferred observation whereas my uh beaver lodge is a preferred observation but what goes on in uh human niche construction and uh because it's the main area environmental engineering is not so straightforward so um for example highly refined food it seems to be a naturally to us like a preferred observation because it tastes good and so it makes life less strenuous and more stimulating so if they're quite if the if the if the company making highly processed food ready meals and so on it's it's acting within quality assurance so it's doing in my opinion active inference realize the free energy principle but what its output it subverts the free energy principle the externality it subverts the free energy principle because what it's offering to us as consumers it seems to be consistent with what revolved to do so it makes life less strenuous and more more stimulating but actually it's not good for our health in the long term of course now and then no problem but if that's all you if you live in a food desert if you live in a food desert we know there are lots of studies that show that unfortunately this isn't good for health so if you take one entity at a time it can be conforming with the three energy principle um through active inference its output subverts the free energy principle because it makes our profit our our preferred observations instead of being good for us not good for us but we only discover 20 30 years later so that's quite a lot to i have a diagram in um there's a simple diagram of this in get paper so the paper's called yeah it's called multi-scale free energy and analysis of human ecosystem okay and there's uh one diagram in it figure one so it's an entropy as well yeah shall i share my screen sure now remind me there it is isn't it all right are you saying multiscoping free energy analysis if you make your system engineering yep all right so here's this diagram then um there it is so what what the so what active inference is leading us to is that from nature is that our preferred observation supports survival an unwanted surprise so an unexpected difference a prediction of what we're going to experience based on our internal generative model so an unwanted surprise so if the beaver's going back to its lodge it's thinking it's going to have an unhindered path back to its lodge and there's the unwanted surprise of the hungry lynx standing in the way and it threatens survival so that's natural free energy principle through um human environmental engineering niche construction within that preferred observations can threaten survival so if my preferred observation is a comfy sofa a large tv screen and a device to change channels and do uh gaming and then i can just order some food in it comes uh very tasty but perhaps not good for me i could be sitting down for 16 hours and then go go lie down for the remaining eight hours maybe all my preferred observations but they're not good for me whereas the unwanted surprise that supports survival could be messages that okay we should not sit down for more than 20 minutes we should take lots of physical activity we should eat make our own food what's going on in human endeavors which may individually conform to realization the free energy principle through active inference can end up subverting outside of that productive act of that production activity and here there's some numbers okay so so beaver could develop a new genitive all right let's see here all right so so in nature for a beaver the the um preference distribution for sensory inputs is for example plus 10 for the protective pond the lodge zero for woodland and minus 10 for a hungry predator but then if there's a bit of pollution you change like okay so now it is getting more complicated so there's plus 10 productive bonding clear water plus two predicted upon polluted water woodland zero hungry predators still minus 10 but then um or the example in this paper is in um some uh island nations very small island nations where there's um a lot of important cheap calorie rich nutrient poor food and then the preference distribution could be plus 10 imported food plus two local food minus 10 no food even though there's clear evidence that the imported food is not good so there it is okay so did that um did that lit up the hands i i wrote down some stuff as well but blue welcome um go for it and then afterwards we'll be stephen thanks so i i was really curious about that subverted um model and like where you're maybe not optimizing like where where what your preference is is not optimizing for survival and i think that um you know you used it in the human context but i can think of many contexts where animals do a similar thing like a moth will fly into a flame and like a cat like i have these two kittens now in my house and they're like such trouble right like their curiosity like had my cat trapped in a box earlier today like i mean trapped like under furniture in the box which was like so bizarre so i mean i can think of of models where it's always not optimal and so is this something that you think excuse me what was the last thing you said it just something that you frame as uniquely human because i can think of many instances of like non-human uh creatures that also have less than optimal preferences um no i'm not framing is only human but what i'm pointing out is is that so for example the moth flies towards the flame but where did the flame come from and the cat's curiosity luckily did not kill your cats that's good but they were going into boxes but where did the boxes come from are these things of course there are misjudgments so all living things can they could be uh and of course with predators and prey that for sure the predators are trying to get the prey aren't they and people can have um and living things can misjudge the how far away the prey is and so on and signaling systems can can get messed up and things but with human beings the scale and the determination activities is is much larger so there's an american author who's his latest book i mean he's called hooked but uh he's studied the the for example the food industry for many years and um he his view is that there's nothing at all malicious going on but it's just if you're if you're in the in the uh a certain business you want you're trying to get as many customers as possible and you're trying to retain them with no no bad intentions at all it's all good intentions then the effects we discover it can be decades later that this is what we thought was is not so good whereas in nature if the the wherever the flame suppose it was a flame from somewhere in nature it was a natural fire well the the natural fire maybe came from bolt of lightning or something wasn't a determined effort over many decades of ever increasing refinement of a market offering but uh so there's more so what i'm showing in the paper that you read that diagram about how the quality management system is like a general it's like an active inference system to reduce free free energy it's optimum for the organization but it can be sub-optimal outside the organization so i think a general word to them for this is externalities just uh two quick things one was trying to explore that distinction between what is human other than the fact that we care about us and each other what is human and i was just thinking of the ants foraging out there in the desert and they don't do margin trading or um staking of their seeds to try to get fractional seeds and lending of seeds it's like layer one seed gathering versus the higher order symbolic risks that the humans engage in so that and the niche modification and the scale among other factors and then also what you said about misjudgment and the ability to misjudge because a lot of people think oh well you know if um let's just say traditional understandings of evolution are about fitness then why why do organisms fail why do all organisms not just maximize their fitness or if active inference is about reduction of uncertainty or using action and inference to reduce uncertainty then why do organisms fail and that's where the engineering perspective comes into play like things just do fall apart and just because there is an imperative whether to maximize fitness or to minimize uncertainty the point is the ensemble there's going to be a distribution of performance like 100 engines they're going to have some range of failure times the ld50 for some pharmacological experiment so just because the framework has sort of a um imperative weather fitness or uncertainty reduction doesn't mean that it's utopian or like pangolossian it means that every single thing always works perfectly so just cool points that you raised but it but also blue um my my uh i have imperfect knowledge that's for and i wouldn't make um i i i i don't know enough and maybe nobody does um like what was just said about ants of course those words were coming for an expert and i don't i know next to nothing about that and many of the behavior of many of or animals i don't know so for sure it's not it's not clear-cut it's not you know one that humans doing this and all other animals aren't doing it thanks steven celet and then of course anyone who's watching live i think this raises some useful areas where we have spaces where it's very free energy minimizing as the main kind of way of getting a handle on things and the broader active inference so for instance when we're dealing with equilibrium chemistry we're dealing with things which are often boiled or heated or made to be effectively dead which is what happens in most engineering processes in most food processes i mean the main thing that coca-cola does by putting sugar in is it makes a preservative someone's saying to me oh why can't they just use water it's an awful lot easier to distribute coca-cola than to distribute bottled water because coca-cola doesn't go off because it's got so much acid and sugar in it it won't go off right so that what you've got is you've got these processes like you're talking about with tqm and i worked in the pigment manufacturing industry so we they were bringing in the iso standards there around pigment manufacturing because they were inheriting a lot of these formulas from like 100 years previous and it was all about minimizing and to make things reproducible and within tolerance but you're always dealing with you never want things to be organic you don't want things to sort to grow in any place in the in the process you want it to all be i suppose you could say dead right but then what's really interesting as you point out is okay that might be what happens within the containment of the factory or the industrial process but it will go back into the world just like it will when it's out now as a product it will be used in an ecological niche and it will not be a dead niche you know that food that maybe isn't even biologically feasible or some of those chemicals are not there's no biological route to create them there's no way to create fluoride compounds biologically it just can't be done it has to only be done synthetically so what's your thoughts about that sticking with minimizing free energy more when you've only got dead equilibrium systems and then when it and what and how much active inference is involved in in that i suppose a first order is only a second or third order and then what how things pop out into the niche in the community or whatever um i think this is um open question i mean at one level maybe it's um in world systems theory this uh the the framing is that there are so now it's in about whole countries rather than just um a company making something in one factory that there are inevitably core i think the terminology is core countries that so it was the netherlands then britain then the usa now going towards china and then there are peripheral countries two levels of those and um the argument is that the the development of the core countries is dependent upon taking from the the peripheral countries and is this entropy being shipped out of the uh so if if you think of like a a factory as a heat pump and it's uh you're getting all the entropy out of the factory operations but then it's where's it going is it being shipped out of the factory it is entry but you've been shipped out of the pumped out of the core countries into the rest of the world where things are confusing so i don't know i mean that's a bit of a a broad sweeping statement isn't it so but maybe on one level that's what's going on if there's if you if you have supply chains value chains you push all the physical disorder out of those by eliminating all the information uncertainty then does that mean that the physical disorder has inevitably got to go somewhere else i don't i don't know so uh but maybe and then but uh but for sure there are externalities and the interaction between levels is uh because in inhuman endeavors it's uh it's a shifting matrix it's not just a simple hierarchy can i just ask one thing just bouncing back off that is i think it's an interesting so in some ways by by things sort of being taken out of the ecological niche effectively by putting in a process a processing plant where it's effectively dead you know you you heat up your sugar your sugar cane raw material in a vat with sulfuric acid and it you know it's all it's all dead it can't it's basically a chemical process that gets put into whatever foods that someone's eating the consequence of having that dead equilibrium based process rather non-equilibrium steady state process is that you're saying it puts a burden back on the ecosystem to try and reintegrate it so in some ways is it pushing entropy out or is it just placing a burden on system to try and find a way to cope with something that it can't naturally incorporate easily into its niche what do you think blue yeah i'm not sure really i'm not um i'm not uh entirely certain on this on this um transfer of disorder it's definitely very important like for example when people say well look at the miles per gallon of this car and then okay but where did the metal come from okay but you know everything is tied to everything and how we evaluate policies ultimately and choices are critical and we do have to understand the relationships of um the world's supply chains but then i thought well it's not like there's a like we're in a bomb calorimeter or a heat bath where there's a constant um number or proportion of some kind of uh negative activity like there could be an ant colony with of in a niche with very low nest mate aggression or bodies like there's organisms with low rates of cancer or in certain niches you know it's not like a statement of them across all patterns and times um but then again it was like so things do fail and there's no perfect state but then at the same time it's not zero sum in the informational game it's not like we're trying to because we're getting energy coming in so can we kind of transmute physical order with information disorder do we have to like offload something to another place physically does it bump up against any of the limits or calculations or density of processing or storage will we not be able to remember enough and then there's a catastrophic failure i mean there's so many components to your questions so i hope that we can all keep thinking about that because it's important at the level of like activity with the supply chain and that's also an important long range question and if we don't have a long uh range preference then we're not going to get there can i have one more bit in there without yeah i think what's interesting as well is there's this question of because i don't really believe there is strictly speaking neg entropy as such so in terms of um it's always kind of a relative thing but it's a useful it can be useful to think about it that way sometimes but ultimately it's always a change so i suppose the question comes in in terms of how much is it around the ability to infer using entropy as much as it is about how you know how much is it to be able to use change in entropy or dynamics of entropy which can be a complex jungle niche you know it could be a lot of entropy change and a lot of stuff going on but it can be very healthy how much is that ability dynamically get at the variational free energy really important for life um whereas for us it's all about you know at times in an engineering context it's just reducing free energy per se as a kind of an overall entropy term which could be actually not so much about variational change it's just an overall general reduction in entropy i wonder what what your thoughts are on that one thought on that on the biological side is what's an industrial engineering relationship ecosystem engineering relationship that has persisted for billions of years or millions of years or just long time however you want to see it the nitrogen cycle and some of the redox cycles in the bacterial relationships biofilms the mitochondria and the eukaryotes now that's a shifting interface but on the other hand eukaryotes can totally count on that interface the powerhouse of the cell as they say to persist but then there's this other sense in which there is still a game theory and a failure mode and uh probably an interface of maximal uncertainty and the requirement for active sensing and trade-off costs so it's again it's not like things i think work 99 of time or 100 or 20 percent of the time that is kind of how the chess board is set up for the for the for it to even be a winning board at all there's just niches where it can't work like there's temperatures that just the population will just die at period so is our engineering condition such that there is a path that's going to lead to what standard of life with what precision and uncertainty for what distribution of people like did we way way overshoot and now it's going to change back down to a different way or is there some critical path changing the distribution of industry because it's not going to be any way it was from zero to 2021 it's not going to be that way next year are you do you are you still seeing my screen oh i'm not could you reshare it okay all right this is a can i reach you do that again this is a are you seeing now figure one better ecosystems in the anthropocene now i see it yep okay yes this is a paper i'll go to the title um nearly awesome or anything but this is um uh not by me by others nothing to do with me i'm just the reader of it um i know one thing that they're saying that you can just so here this this is the same these are showing the same things but a is where human impacts on migration-based matter ecosystem it's it's just sort of implicit and on this framing of it they say that okay so we explicitly deal with the uh socio-ecological framework about okay so what are human beings what mental models of human beings got what regulations arise from that and that inputs into it so let me go to the top so you see the title yeah embedding method on embedding meta ecosystems into a socio-ecological framework and this is a response this is just a short paper it's a response to somebody who replied to the no it isn't actually nope i'm mixing up with another one there it is that's the title anyway so i'm wondering if um that may be that each each way of um making things has got its own entropy dynamics and the in this entropy and dynamics that there's physical disorder is reduced information uncertainty is reduced but maybe that then pushes the entropy out elsewhere and then just doing more of the same thing it's just not possible there's a limit to it and with world um systems theory this is the breaking gist of it that there are these core countries and there are peripheral countries that serve them maybe the if there's one way of doing things you can't address the entropy that's caused by that by trying to do more of the same or trying to tweak that way of doing things and maybe something quite different has to be done and maybe there's something about something to be learned from landscape restoration ecology in this about that about um habitat fragmentation that that okay so here are these habitats and they've grown and they're thriving but these ones over here they fragmented they split and this there's nothing from them but dispersal so as people are dispersing from parts of the world that are suffering from premature deindustrialization maybe there's something in entry by dynamics that could be that well that this doing this type of industrial production this it drives out the entropy but it's shipping it out it's going out elsewhere and the entropy that is then um in other places it can't be dealt with other than by some in another way of existing does that sound in any way sensible a very interesting area multiple levels of nested modeling and uncertainty and how informational and thermodynamic aspects transmute into one another and arbitrage with one another or relate to each other so it's really one kind of thought on that and again to the fast food and things that might be unhealthy for us or policies that might be unhealthy in the long term for an organization it it made me think about being on a road trip and knowing that there was going to be not just the hyper stimulating experience which has been argued in many other frameworks like in reward based frameworks which is what you know dopamine the reward molecule reinforcement learning reward learning they'll already have a framing for how those products hyper stimulating lead to failure modes of complex societies via hyperstimulation okay so that is sort of a common stance with active inference we can also kind of highlight the fact that those products have lower variability and that there's actually very bland foods that for reason of blandness that people or like a tv show or something that for reasons of narrative or aesthetic just um similarity people will re-watch i mean even to videos that of types of people didn't expect why do people watch the log burning or the goop playing because they're sort of like the um frontier of optimal foraging is about like how it's going to sound which is a very small thing to be resolving relative to the more um deeper unsettling levels of information that could be revealed like someone else's perspective of you or something like that so it's just interesting how we can actually as you point out like the supply chains are modeled on one hand by these cyber physical digital twin frameworks that need a unifying approach and so on one hand active inference does provide that but then looking at the other side of the equation so what you didn't discuss in our paper for these weeks but where you left off when you start introducing the biological and the survival imperatives then we can also have some new ways of thinking about how those industrial products through precision hacking rather than reward hacking end up leading to maladaptive behavior with or without reward incentivization are you um what are you seeing on are you seeing my screen now a different yep very different diagrams you see birds yeah generalist bird gap specialist butterfly interior habitat but especially speedless so this is another paper um i'll again i'll scroll up to the top you see this tile um and it's making this point that uh landscape composition and configuration go to different species perceptions and so the first one's a human and then different species and as it's neatly illustrated there that the same the same place looks different to different species and so what i'm the the kind of the terminology here is that there are patches habitat patches and they can be corridors between them or stepping stones between them and the stuff the other stuff is called matrix and i personally can't ever see that word or say it without thinking about the movie so that is a bit of a problem but uh but in the landscape ecology is called the matrix so this i think this could be useful of illustrating that well okay that so here's the road to the mine these uh front country come companies built the road to the mine they built the mine and they're taking stuff out and they're taking it back to their country to do highly efficient manufacturing with it and then they send their completed goods back and but then what's left so if you just measure that in terms of gdp growth then the gdp goes up divided by the number of people and it appears that prosperity is rising but if if you look at it in this kind of view it shows that well there's a lot of blank spaces left to be left here if we think about it in terms of this uh restoration or regeneration it's looking at it all of these because of course the the perspective of the human the generalist bird the gap specialist butterfly and the beetle they're all equally valid and um you know what this paper's called landscape ecology and restoration processes and even within the same physical overlap one bug might be underground or in the roots or on the ground in the canopy so even the physical micro niche that different strategies or individuals experience in the same spot in the matrix can be very different so there's just such richness to this ecological thinking it's just challenging to bring it into play in some of our cognitive endeavors but i think that is also changing rapidly so i've taken um an ecological perspective in we're doing one paper on meme ecosystems and rhetoric ecosystems by the end of this year but it's not active direct maybe it'll have some relationships it could well do um are you now seeing a table to number one all right so what i've done here is i've done this from a uh ecological perspective and well you can see these terms but one thing in uh restoration ecosystem restoration is mobile linking organisms that like birds and animals that can travel quite a long way they can go so after this is a forest fire animals that can travel quite a long way particularly birds they can come they can fly and drop seeds and and things like that so this is an established concept mobile linking organisms and i'm pointing out that this is something that we've done in mobile factories and then energetics another key issue is that if there's no if all the energy if all the matter and energy are flowing then life isn't sustainable and uh of course there we are so i've i've done that relating to my uh favorite subject movable production technologies so that takes i think this is decentralized systems isn't it that the second point that was it that was the lead in the segway okay nice how about oh go for it um stephen yeah so i think this is the if everything's so at the moment there's this uh centralization oh let's have a look at this i've got um i go again so let's have a look here what do you see now do you see a taxonomy uh taxonomy and manufacturing distributions and their comparative relations to sustainability here they are there are distributed distributions and centralized and as we see there's plenty of distributed options and uh then i compare them in terms of economic ecological social and institutional sustainability and where i think active inference i've just started to look into this that this agent environment systems at framing is the the agent has a generative model but the world is a generative process the sensory inputs are coming from the generative process to the world i think that this enables um active inference to be linked to ecosystem dynamics that what we've been well what i've been saying before about this centralized distribution it um it's super efficient and it's getting uh engineering out physical disorder and information uncertainty but maybe that entropy is going somewhere else and uh new information is needed to activate the more distributed options so do you think do you think that um if the only if only a certain um so if if there's a predominant paradigm which in this case there is industrial based on the 300 years old paradigm the information about how to make things is all about that and how to improve then does it become a zero-sum game if if so if it is shipping out um entropy like a heat pump out to the peripheral nations do you think it's a zero-sum game if there is an alternative information so if nobody knows or there hasn't yet been uh conceived an alternative then then is it like a closed system so in a closed system has if it's pushed out of one place it goes to the other places so is it is a cl is it a closed system is it a closed system if all the information is about one paradigm so is this like uh for example does am i in any way coherent here it's an extremely important and um you know socio-technical systems political systems tokenomics that last forever are these the the perpetual motion machines of the 2020s and just a new level of uh computational uh detail and flourish but the same kinds of fallacies and then whatever the thermodynamics of the earth our niche is finite and we've now made some potentially whether or not the sum is zero or even negative or could be positive but is negative there may be informational and strategic games that um don't resolve like heat baths they can't be understood with some of the at least most tangible metaphors that thankfully guide the evolution of theories that our first principle like fep like carl fristen with the wood lice seeing the trajectory of a real thing and that's based upon health and um longevity blue yeah i had a question about the table that you pulled up earlier um specifically about ecosystem engineering and so i'm not sure if it's just like a term that i'm not familiar with or like in in terms of engineering i'm kind of naive when you're thinking about ecosystem engineering um i i i equate that to niche modification or like designing a system like even you know you could design like a world system here it says it does not require overriding the natural evolutionary balance of equal equal ecological fitness but like i think about like the human paradigm blue could i just could it just happen but that's movable production saying there is that if you've got a movable factory so a truck if you've got a secondhand truck with a secondhand container on the back of it it does not require overriding the natural evolutionary balance it's it's specifically in relation to if you look at the top it says movable production technologies so so how is that ecosystem engineering though like i mean no no it isn't yet but the point uh obviously this table is not clear and it is my responsibility it's not clear i am responsible for this but so the the what this meant by this is because further up in the paper it's pointing out that if you have fixed so if you've got a big fixed factory you build a load of roads huge roads going to it and so on that involves a great deal of ecosystem engineering a mob a mobile factory it can just go over rough terrain also to uh play devil's advocate against the author the enabling function a uh realization of something something physical that can be an ecosystem engineering agent ak niche modifying agent um that it's gonna have to be a niche modifying agent so this is just sort of then connecting that to mobile organisms so you have quote plants aka factories that are planted and then those can be like they're not like trees simply but they're planted and so they have strategies not dissimilar from other rooted organisms but then you have mobile organ linking organisms so it sort of connects that all the things that mobile like all animals do and can do with what a factory could do ecologically so it just kind of makes me think about like what kind of biological organisms take their niche with them like maybe a snail or a turtle like you know to take take a large component of like their habitat essentially with them wherever they go the mobile technology made me think of that yep some people like to travel with uh light you know just the keys or something other people full backpack every time stephen i like this idea of something coming out of the factory going into the space and then things being secondhand a third fourth fifth hand because there is something there that is delivered is this i mean if you're going to use entropy i suppose you could say low entropy kind of nuke new vehicle you know whatever it is and slowly it gets incorporated into the ecosystem it starts to get scratches it starts to get this and you see in india you know people start decorating the vehicle you know the all this sort of stuff will happen over as it becomes somehow incorporated into the niche i suppose you could argue so i think it's quite quite interesting how that maybe starts to change the dynamic i i'm still not sure whether entropy is so useful term or whether it's some other way of looking at a metric maybe something to do with free energy minimization as a as a thing but um better yeah one other piece of that table that was interesting was there was multiple goals like sustainability and fitness it uh is not a univariate optimization we're not just doing reward or utility maximization we can choose multiple goals and we can have multiple human goals which is especially important for things that you know we care about but then this one of mutualism is interesting because a lot of times mutualism could be seen as like an outcome like the outcome of these two bacteria's relationship in the niche is a commensalism or mutualism or symbiosis whatever it happens to be but actually using this idea of the the outputs like the observation states of a higher level bayesian model being like the priors on a lower level nested model having a goal of mutualism is implicit in the lower level model enacting that mutualism because you're not accidentally going to metabolically coordinate through millions of generations and so this is kind of an interesting way where uh like we have to want to make it work because we're not going to accidentally make it work and get along but that can be a hyper prior like a belief about how things ought to be or how we want or prefer things to be and then we enact policy along those lines and by the way i do um other uh notes it's a by the way in um in another paper i do point out there are limitations to movable factories so i'm not uh saying that this is a panacea but it's sort of like like let's just say um cactus versus like a desert animal you know the desert animal they have different ways of getting water and so but when we can design the system we can do trade-offs like you could have something that's stationary but then isn't just holding all of its own money because its money can be like on a digital system so it can have some of the advantages of a stationary organism from an ecological perspective like a tree but then like the value of the tree um could be protected in some other way because we have some other affordance and or that forest could be de-risked at a higher level because it's like oh each one of these is we expect to fail this many percent of the time or we have these quality control checks on it yeah i think a lot of thought is required about about these things isn't it it's i don't think there's one big answer but but at least i think one good thing about the energy principle active imprints that it ties into action and that ties into energy this is good yep when you're doing the thought experiment pure uh gedenkin then it's like can be elucidating you know imagine you're going towards the speed of light or imagine you have a a physical mass of this size but then there's no constraint which is again double-edged sword on some of these attributes that require action so that's that's an interesting idea that it ties us to active states which is pragmatic and useful but also interesting and then also one thing on the affective system and then blue you can ask anything was just we've talked also about higher order variables in organisms being kind of like affective states whether they were the valence like positive or negative or the anxiety or the relative beliefs of an organism um how do we design for when there's higher order states maybe even ones that could be considered affective either for people or for organisms or i mean organizations that have this kind of organismal structure but blue so interestingly like this is going to touch on what you just said but also going back to what you said about mutualism and the goal like has to be mutualism so so i don't know um i really have to uh contact the author of the um 2020 paper uh sims is his name how to count biological minds symbiosis the fep and reciprocal multi-scale integration which i found um very fascinating but i would love to invite them on for a stream someday uh the the whole point of that is that there's like nested cognition happening at the biological scale and specifically that paper looked at the squid and the vibrio fisheries that like light up the squid in the in the light organ you know in that kind of context as in an industrial organization there's like this nested cognition happening um and where the point of that was the goal wasn't necessarily um mutualism or in that case symbiosis that that's not necessarily the goal but a bigger grander organism was constructed out of the union between these two organisms and so perhaps in an industrial scale that's also something that happens right i mean there's different parts to the organization that'll keep it running and you factor in all the supply chain things and and so there's this nested hierarchical thing that happens and something bigger that has its own kind of cognition occurs as a result of this like mutual uh beneficial relationship we see the paper the paper we see the paper that stephen just put up yeah there's some i thought it said in here so i thought it said in here somewhere about yeah these different types of relationships so mutualism is beneficial beneficial parasitism is beneficial harmful but in between those there's beneficial and neutral there are several in between mutual and para parasitism there are several other states and i just remember that the one of them is beneficial and neutral if i could give a kind of random thought on that and why the informational sphere might have a different entropy dynamic than the physical the electromagnetic spectrum that can have a lot of positive and neutral relationships like if someone's operating on two different wavelengths and one is only on one wavelength then they can be like of mutual service to each other or one could just be totally non-interacting because they can be overlapping in space and time at once and one person might not even have to be aware of it whereas in the tangible i guess most embodied one can be you know only one thing per spot that's zero sum there just can't be two two atoms in the same spot maybe you know two laws playing out in one territory but informationally there's probably a lot more multi-channeling and that might give enough room for practical success if not ultimate theoretical answerings stephen slit and then blue this brings up a lot of issues it's very fascinating um i'm i'm curious um so coming back to this idea and i i mean daniel mentioned hyper priors i things that we kind of have fixed um i actually looked that up actually to make sure i understood it but things that we have is very distinctive and i'm again trying to compare these these types of um environments when one where you've got an agent wanting or being able to just calibrate in amongst the noise and ones where you want to just keep it as clean you know so and and the challenge is sometimes when you have a quality management system in those places so for instance i suppose one thing that comes to mind is the experiment with a rat that they used to have and they've had for years about having cocaine or addictive substances and how they would they only needed to have a couple of times and they would be addicted and it was seen as this kind of proof of principle about addiction and how it works and then the more recent work where they instead of putting them in a bare box and just having the cocaine or whatever they were being um sort of exposed to they were in a more ecological boxed you know where there was things to do and they suddenly found that the behavior of the animal was very different they wouldn't necessarily become addicted in the same way because they would actually because they were in a better ecology now in theory the entropy's gone up there if you're looking entropically but for the animal they've been able to engage and and dynamically process that that's very different to how you would try and create you know a medical production line right you you want to keep it clean you want to keep it sterile um you want to have those sorts of so i'm just curious about that that and then the danger maybe in the biological system of this hyper prior say the some sort of kick or something that comes from food not having any other compliments around like like as daniel you know so there isn't other things so that just naturally kicks in like the reward goal which may be always there as a default if there's not other things to complement it it just it starts running the show and that could be part of how we get into these maladjusted behaviors that you were talking about i wonder what your thoughts are on that just quick point of clarification that's not what a hyper priority is that's not a hyper-stimulating stimuli a hyperprior is a is a prior over another prior in a bayesian multi-level framework so it doesn't have to be accentuating or differentiating or extreme or maladaptive it's a general framework for the relationship between variables yeah well i was thinking it was more the hyper prior could give a it's a it's a fairly fixed priority and it's one which is a go-to solid prior and it's useful for being that does just saying having that present hyper fire could be fixed or learned okay okay but it tends to be hyper prior means it's quite accurate isn't it it's quite it's not fuzzy hyper is i think a spatial designation like supra not in the extreme case hyper is only referring to the fact that it's a a parameter about another parameter so it doesn't tend to be hot or cold or anything it's just the general phrasing okay got you yeah so but stephen fox if you want to add anything otherwise okay we see the head yeah so now the effective system here here's my diagram so in in this um saying that uh it's a little bit so small i can't see it myself but if there's the same starting point but um of a generative model so if the human being in human robot interaction knows that um okay so i i'm i'm in the predictor i'm human i'm in the predictive processing business that's what i'm doing and um um so i've got prior beliefs and i have got generative models robot can be um what does that say personas and scenarios for robot based on the humans uh well that's world view but generative model and human readable model of the robot mind that corresponds to that that's um and this is in aligning human psychomotor characteristics with robot an augmented reality in this there's a bit of a model there that's also very related to the felix shoulders paper trust us extended control active inference and user feedback during human robot collaboration ooh i'll put it in the youtube channel i'll put in youtube chat but brilliant go for this one so that's um it's all in the past experiences physiology so this is body memory in there personality gender culture emotion reasoning in biocybernetics so this paper i'm relating that to what how this can affect interact with robots exoskeletons and augmented reality we got here a few equations there it is but here the two i point out in this that um you find in previous production literature the uncertainty of human behavior this is this is true so in previous production literature the uncertainty of human behavior has been recognized as a source of productivity quality and safety problems however fundamental reasons for the uncertainty of human behavior have been framed previously as a black box so what i was trying to do there was relate it to to open up the black box frame it as an open box fundamental reasons can be aligned with production technology to facilitate improved production performance so that's as far as i've got with that it's almost like for the reward utility driven person you can say there's going to be an increased expectation of performance for the uncertainty minded you can say there'll be a reduced variance for the resilience minded or for the reconfiguration minded that there'd be an increased capacity to rapidly reorganize so i think that we'll have all or none so my thoughts kind of dated at this point but just going back to um touching back on to mutualism and um you know these these the electromagnetic spectrum when you're talking about how like you know the signal can amplify there's also like the possibility for the signal to cancel right if if the you know wavelengths are are the same and they're heading in opposite directions it they just directly cancel each other out and it kind of made me think of like cancel culture and also like the relationship between um you know the parasite is is also like a nested biological relationship um even though it's not necessarily beneficial but there's still this nested layer like the parasite is unable to function without the host of course the host embodying the parasite like is a completely different cognitive cognating creature and i think about like the zombie ants right that um i don't know if you guys have heard daniel can probably unpack the zombie a lot a lot better than i can but it's like a parasite that then drives the behavior of of the ant due to some like rewiring of the cognitive process so even in in the parasite instance there's still a greater cognizing structure going back to neutralism again the the other terms are so commensalism is where one gains benefit while the others it hasn't there's a new neutral no benefit no harm and then there's a mentalism where one is harmed or the other is unaffected so they're in between so there's first benefit benefit then commensalism where there is uh benefit and then there's a mentalism where harm and neutral and then parasitism where there's one is harmed and the others benefits i i wonder if this has to do with the movement from maybe just its semantic but from sustainability to regenerative because sustainability it's like the sustain on the guitar sound of the bell like let's sustain let's have this car and sustain it or make it it can be meant to mean more than that but it's about sustaining something that exists versus regenerating means like we're gonna have to be doing more than just persisting or stabilizing it that it has to be also actively having this process of uh of regeneration which always involves like multiple parts like bucky fuller saying that unity is plural and at minimum two so then now thinking about how from polyculture of growing crops to monoculture the green revolution and everything and then now oh maybe there should be the fish in between the roots of the plants and then the birds and the mushrooms like this are these stable ecosystems that is what remains to be seen but some human modified niches like sourdough bread and things and beer have been really successful so maybe there's higher order combinations too that could be really successful and be very productive but this is kind of a framework that helps us at least explore that so that's quite cool i was at um remotely of course it was there was a launch the south african um grassroots innovation programme they did um a launch yesterday with five lady innovators and one of them was doing um movable aquaponics she seemed to have addressed it's quite a sophisticated system but it's all uh it's all set up in buckets that can be carried one person can carry the buckets that's the size they are and then you build up the farm by different just having the number of the different the more buckets you get the bigger the aquaponic farm so that was interesting and then again with the physical zero sum and then the electromagnetic or informational or financial non-zero sum if the overlapping is just given the tools that already exist in the ecosystem that you find yourself in like your dropped on survivor island you still may be able to find some synergies most of them you won't know but they're over long time traditional ecological knowledge can learn about some medicinal combinations and oh these two should be combined for the boat and these kinds of things but design and this is related to what steven said about the product being plugged into another external niche and what you wrote about the cupboard being attached to the wall like we could design some pretty minimal pieces that might really scaffold you build a trellis and then something grows within it so the engineering might not be huge to connect a few pieces that have a latent capacity to work together they the word you use their scaffolding has that got any of course i've heard it before but does that have any basis in nature to your knowledge i would certainly say so um from the spicule structures like in bones and the way that um physical structures are scaffolded to um embryogenesis and the developmental scaffold scaffolding of the uterus and core caretakers and different insects and humans and the family and so i think there's the cultural scaffolding and then all the way down to the microstructural cytoskeletal scaffold for scaffolding enzyme complexes that then can do very efficient things too yeah it's interesting because the problem with the the challenge with um so-called grassroots entrepreneurship and innovation is scaling it up that's the the challenge that's that's what was said yesterday but that's what said in india there's this problem that solutions are developed and they they work but they at best stay where they've been in the village where it's been developed it sort of just stays there and so scaling up and it's a um open source um would it's well there it is the technology is there isn't an open source ecology it's uh it's possible to put all engineering designs and builds and materials methods of work and all on the internet where anybody can access it but still things are not scaling up thanks blue so i think about scaffolding um in terms like neural systems but also in societal systems in the brain like my phd work was focused on finding a scaffold that will promote the growth of neurons across um across the lesion and so when you have a traumatic brain injury the glia which is like the natural scaffold like freak out and form a scar to prevent like the whole rest of the brain from dying because cell death can be like this kind of cascading process and so so the glia and and it literally results in like a hole in the brain where the injury was so i was looking at these bioengineered scaffolds that promote the growth of neurons across that hole so so like it'll it'll form a bridge or a structure scaffold for the neurons to grow um and then you know in terms of the the societal um setting i think about um scaffolding as infrastructure and and i mean even in an organization like you have to have bridges and roads and internet connections and utilities and you know these things are it's the the infrastructure that's necessary for kind of new branches or systems start up and even in like the farming terms it's like you know the the um like i don't know here we just call them the ditches but like the irrigation canals i guess is probably the proper term like that you know the um where you can get a river to flow off into different subsections and then irrigate the farmland with that water so so i think about that as as the scaffold awesome blue nice connection and sounds like cool phd research we see your screen stephen so you see grassroots innovation program yes okay yeah so that was that was yesterday um although the press release was in july but it's not it's not detailing the book but there we are this is the um so in south africa there were the rate of unemployment is i think officially it's more than 30 percent but um unofficially it could be as high as 50 so they're now trying to address at least partly through this grassroots innovation problem program so one and there are there are similar and we know that the book decades ago small is beautiful gandhi was before that was promoting this but it does it's not scaling and so there are these in between industrial production there's just this ever-increasing wasteland but efforts are going on around the world and grassroots is a good term because it's just growing from evolving naturally doesn't it sounds like really interesting work thanks for sharing it stephen celet yeah actually that ties in i actually was running a project in south africa funny enough um and working on looking at how to create community driven innovation through strategic strategic conversations at the community level and there is there is an interesting difference between grassroots i think and community driven so i think that grassroots is often kind of um tied to it's like the base level of the organizational structures you often hear it with political campaigns and stuff is the grassroots but those grassroots are the grassroots of a campaign where people sort of get into there which is a little bit different to the kind of a community embedded community driven scale and that comes back to the scaling question which is the same thing that had the problems that so they're scaling there's now when you talk about scaling up and that's a big challenge is how does stuff get scaled up and they're often now talking as well scaling out if you have a program or scaling deep you know some things might scale up because they go up to a higher level and then they basically get dropped down again but how do these things get scaled out there may be some i think you're you're you're tapping into a few multi-level issues which i think are quite quite important so i wonder what your thoughts are about going at uh going into these kind of embedded local context to create new seeds so to speak for like adaptability and innovation it's like a seed is a biological planting metaphor the scaffold is a um highlighting the industrial component of a biological process like you still need the tomato seed so it's not saying it's not community driven it's just like a soil metaphor like how these communicate what our vision is and they really do matter and so they're really um important points actually here's a comment from kevin in the chat who wrote is the scaffolding a schema or is this supposed to be anatomical that is is there some physiological analog in the human nervous systems or genome that corresponds to the scaffolding i think blue made an awesome point about the glia and about connective tissue being an immune tissue being scaffolding physically for like the synapse and for cells and then the genome oh yes the chromatin scaffolding goes deep in the nuclear pore structure and that's a whole cool area and maybe there'll be a day for transcription factor active inference soon enough so um what you're saying these hands this is a an organization um active in botswana and everything they're doing there there we are you can see yep so this is led by uh thurbison he's um applying design methods in um uh communities in southern africa with great effect and this is all uh so the idea is that people are with their own hands are developing innovations and as you say you can be scaled out for example in in that part of africa that there are many movements like uh many initiatives going on but it would be very beneficial if there was um framing a kind of unifying framing of what they're all doing which can be seen as an alternative an alternative form of socio-economic development rather than just as something that's uh well fundamentally inferior to industrialized to to the conventional model of industrialization so all suggestions are welcome i just put in the youtube chat a volume by capparel grissomer and wimsat developing scaffolds in evolution culture and cognition and so the paradigm change as steven kind of hinted at very early is like from systems where we eradicate the life to where we welcome the life and all that it brings and we don't need to build that and we can't build it we're not going to be able to compose the lipids and the millions of years of fault tolerances into some sort of uh monster that helps our living systems we don't even have to though we just have to build an engineered scaffold and the scaffold can be radically different than what the final product looks like the scaffold can even disintegrate if it's biodegradable it um it's a really interesting area that this discussion went which was like towards the engineering that makes us think of like foundries and some of the technical and precision tying them to some natural processes like this evolution culture cognition volume does and unifying frameworks those uh together almost point a way towards engineering a scaffold not a booster rocket or just a bunker but bunkers might come into play of doing some scaffold design so that there could be reduction of expected free energy through deep time so pretty interesting um point stephen thanks for sharing all these resources too blue just a quick comment um you were talking about the scaffolding disintegrating sometimes the scaffolding is designed to disintegrate like i think about in in biological you know terms we have like webbing between our fingers that enable like the growth of the digits and then you know it's designed to be eroded uh in in the developmental process um so sometimes that's that's part of it or even like when you get a wound sealed you know you have like the the stitches that scaffold that form the scaffolding so the healing process can occur but they dissolve inside the wound actually that's a good that's quite cool to think about that as well in the and uh when when we whether in areas that we live in or areas that we don't live in when we plan to intervene are we planning to make something that sustains itself or are we planning to deliver or co-create a scaffold that is planned to disappear those are really different views on local and distal intervention steven with a screecher what's on the screen stephen fox yes so now this is um this is an initiative called connected hubs so this this is coming from the southern african uh innovation support program uh the idea is to that there are which support innovation and entrepreneurship and these are this particular one is um so as you can see the connectivity is in this part of africa and what we do is build capacities and support so it's connecting uh different hubs together and so a hub it's just what that means is it's just that usually they've got a physical location and they there are people you can contact there remotely and physically you've got can help with developing um ideas for entrepreneurship and social entrepreneurship so this is something that's going on it's connecting hub so maybe there's well hopefully this is the intention is this persists after the southern african innovation support program is on the second phase now after it finishes so this is one i suppose it's summed up in this uh diagram here this idea that there are more and more hubs herbsing and the european commission is um trying this uh africa eu digital innovation hubs oh we'll see this and then there we are so this is kind of things that are going on so i'm involved in this project sustainable um network of digital innovation hubs so that's something that's uh coming down from the african union and the european union and the hope the aim is that this is an uh a somewhat new way of facilitating socioeconomic development and the idea is to try to get auto catalytic effects by you know more connections more um more dynamism well they're always welcome to come talk about active inference at whatever stage in the last few minutes we'll just have steven and then blue yeah just give kind of last thought or question here as we close out okay yeah no thanks very interesting and i suppose with active inference i think very interested in these these developments it's actually how i ended up getting into this because i was trying to develop community-based hubs in south africa funny enough in rural rural maputo land um and i think what's quite interesting is daniel's this this idea of a unifying framework rather than trying to reduce down all the complexity which is what i was doing for far too many years is really amazing and the other thing that all of these aspects that you're taught you've just shared and that ties in with this lab is it's it's about participation as soon as you bring in participatory processes you you bring in these active inference dynamics you bring in this complexity you bring in this need but also the ability to do what mass manufacturing can't do but like i thought you said it's it's seen as being inferior or in our culture and i think seeing it as being vital but also challenging is really cool so i think that's something i'm going to take away because i can't i won't actually get to stay too much longer i've got to jump on another call but that's something i'd like to take away so thanks very much brilliant really really enjoyed today thank you so much thank you steven see you later um so just a couple final thoughts uh john boyk and his idea of connected hubs i mean he was on like probably six different um live streams and uh he gave a very good overview of sustainability um and the you know how to build an architecture toward a sustainable future um so the connected hubs kind of reminded me of that and then um the one other thing was um just about scaffolding and you know how do we build kind of a scaffolding for our cognitive architecture and is this a scaffolding that's maybe designed to dissolve and so i think about you know scaffolding for our cognitive architecture as as education right like we send the kids to k-12 and they you know form this like shared understanding of reality and and two plus two is for and how to read books and so forth and this kind of um education maybe daniel will remember at least in college when i started college um in biology many things are so complicated that they just give you like a very broad overview that doesn't make sense or i was always searching for like but that doesn't make sense but that doesn't make sense like in your entry-level classes and and then as you get like deeper and deeper into studying biology or biochemistry you start to kind of understand electron transport and these different like things that are are very like micro but the macro level doesn't make sense without understanding that that micro and in my mind this this kind of educational or that way that i was taught in college was kind of like building a scaffold that was designed to disappear anyway um but anyway yeah thanks for the for the talks today and for coming on it was if i can get my last comment and then to the author for the final comment i just love this idea of the biochemistry education as the removable scaffold like you learn it once that there's the essential amino acids and then you never wonder like why on the nutrition label is there essential and non-essential you forget the details the side chains you know the c minus on the test that all disappears with something that's generalized but otherwise we would look at macroscopic phenomena like why can't i swim when there's this algae in the lake well there's some molecular answer we could know at least how to sense make in that area might even know the specific details but also it can be removed in a but can it be removed intergenerationally a chemistry to alchemy reversion with similar or different tools that would be quite something so stephen with the um final notes but just wanted to say thanks for coming on these streams and it was just like super interesting discussions well thank you very much it was my pleasure and i look forward to further cooperation with you all great if you or any of your collaborators on these projects or your contacts these hubs wanted to present on this or facilitate in any way then we can talk um another time very good thank you very much thanks again steven thank you blue bye